Jan 10, 2026

AI Project Course: Grading Rubric

High-level Grading Rubric

Component	Weight	Assessment Method
In-Class Participation	20%	Individual assessment during class
Final Presentation	20%	Team presentation with individual components
Project Implementation	60%	Team assessment with individual accountability

Fine-grained Grading Rubrics

1. In-Class Participation (20%) - Individual Grade

Grade	Criteria
Excellent (6.0)	• Actively contributes meaningful insights in every breakout session • Asks thoughtful questions that advance group understanding in lecture or breakout sessions • Completes all in-class exercises • Demonstrates good understanding of concepts
Good (5.0)	• Participates when called upon • Significant attempt to complete the majority of in-class exercises • Shows adequate understanding
Satisfactory (4.0)	• Present and attentive; absences of 4 or more days can negatively impact this grade • Minimal but appropriate contributions in group work • Rudimentary attempts at in-class exercises
Below Expectations (<4.0)	• Passive participation or frequent absence • Limited contribution to group work • Often does not attempt exercises

2. Final Presentation (20%) - Team Grade with Individual Assessment

Weekend 6 Presentation

Criteria (weighted equally)	Excellent (6.0)	Good (5.0)	Satisfactory (4.0)
Technical Accuracy	Technically sound description with appropriate depth; correctly explains ML concepts; detailed responses to Q&A	Mostly accurate with minor issues; good responses to most questions	Some technical inaccuracies; basic responses to Q&A
Business Value Communication	Compelling business case with clear ROI; excellent stakeholder priority alignment	Good business justification; benefits of choices explained clearly	Basic business case presented; some misalignment with stakeholder priorities
Individual Contribution	Clear ownership of presented section	Moderate contributions to presentation	Active in presentation but contributions are unclear

3. Project Implementation (60%) - Team Grade

Format

Documentation: You are free to use whichever platform you feel most comfortable with for creating this documentation. We recommend either a collaborative document editor (e.g., Google Docs), or a README-style Markdown file as part of your GitHub project repository

Code: When you accept the assignment invitation from GitHub Classrooms, a repository for your team will be created. All code should be kept in this repository.

Baseline Requirements (Grade up to 5.0)

Assessed through homework assignments, project code repository and final report

Homework Assessment Method: Live grading sessions with TA for each HW assignment

- TA randomly selects team members to explain specific components
- All team members must demonstrate understanding of the entire solution
- Failure to explain assigned component may result in grade reduction for entire team

Pipeline Assessment Method: Assessment of project code repository at end of semester

- Teaching staff is able to run a team's pipeline using their project's GitHub code repository and query system with new inputs
- Minimum requirements for each project are given in project Briefing Documents

Report Assessment Method: Combined mid-term and final report assessment by lecturers

Your final report should demonstrate the thought processes and iterations that went into building your system. This documentation serves both as a technical record and as evidence of regulatory compliance efforts. Below are the required sections across all projects; additional project-specific documentation requirements are provided in project Briefing Documents.

Regulatory Compliance Assessment

- Document how your system addresses relevant regulatory frameworks (e.g., EU AI Act, GDPR/revFADP, domain-specific regulations)
- Explain where human oversight mechanisms are integrated
- Include risk assessment and mitigation strategies for your specific application domain

Technical Decision Justification

- Document the variables from your dataset that you used; for variables with personal information, give justification as to why these are ok to use
- Describe the evaluation metrics used, why they are appropriate to your domain and which stakeholders they are important to
- Provide comparative analysis of different methods tested (with quantitative results)
- Explain the rationale behind each major architectural choice with supporting evidence; explain the trade-offs considered and why specific approaches were selected

Testing and Validation Protocol

- Summarize generalization results and observations from robustness testing under various conditions (missing data, outliers, edge cases)
- Provide temporal validation if applicable (performance over different time periods)

Lessons Learned and Failed Approaches

- Document approaches that didn't work, including explanation of method and quantitative evidence of failure (metrics, error analysis)
- Describe insights gained that informed successful approaches

Extensions for Additional Credit (Grades 5.0 - 6.0)

Below are some of the directions that would count as extensions to the AI project. Each one adds **up to** 0.5 to your final grade. This list is not comprehensive! If you have questions about whether something would count towards the extension points, contact the lecturers

Extension Category	**Examples**

Advanced Modeling	• Model architectures outside of baseline implementation • Appropriate, project-specific loss functions • Cutting-edge approaches outside of baseline implementation • Extension to additional use cases
In-Depth Analysis	• Comprehensive error analysis and/or failure case studies • Ablation studies • Statistical significance testing • Advanced data visualizations (must be project-relevant aspects of data)
Advanced Interpretability	• Feature interaction analyses • Custom visualization tools, e.g., SHAP prediction attribution dashboard • Automated counterfactual generation
Engineering Excellence	• Comprehensive testing suite • Modularity and extendability of code base • Doc-strings for functions explaining inputs and expected behaviors
Domain Innovation	• Industry-specific metrics • Regulatory compliance features * Identification (or definition) or case-specific bias/fairness metrics
Failed Experiments	• Detailed documentation of approaches that didn't work & clear analysis of why they failed • Provide sound recommendations for future improvements

Extension Assessment Criteria:

The following will be considered when assessing extensions:

- **Motivation & Justification**: Clear explanation of why this extension could improve the product
- **Implementation Quality**: Code quality, documentation, reproducibility
- **Effort & Complexity**: Depth of work, technical challenge addressed
- **Analysis & Insights**: Quality of analysis, even for unsuccessful attempts

Notes

- **Team Accountability**:
    - While work can be divided among team members, everyone is responsible for understanding at least the basic components of the pipeline
    - Random selection during homework grading sessions ensures shared knowledge
    - Consistent failure to explain team work may result in individual grade penalties
- **Failure Documentation**:

- Failed experiments with good documentation receive credit. In the “real world,” this prevents future initiatives from repeating unsuccessful approaches
- Include: hypothesis, methodology, results, analysis of failure, takeaways